

OYYK, Kazakhstan

WMO: 382030

Lat: 43.783N

Lon: 70.933E

Elev: 1201

StdP: 14.07

Time Zone: 6.00 (E06)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.3	(c) -1.4	(d) -13.0	(e) 2.8	(f) -6.1	(g) -6.9	(h) 4.0	(i) -0.2	(j) 22.5	(k) 31.0	(l) 20.1	(m) 31.4	(n) 2.2	(o) N/A	(p) 0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.6	99.1	65.9	96.5	65.0	94.0	64.0	68.3	91.2	66.6	90.4	65.3	89.5	5.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	84.2	76.4	58.5	76.4	74.3	55.8	69.2	74.2	33.2	90.3	31.8	90.5	30.7	89.6	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.9	(b) 15.9	(c) 12.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 20.9	(b) 15.9	(c) 12.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.6	22.4	24.8	39.2	53.8	65.7	75.6	80.4	75.8	64.8	50.2	36.2	28.3		
	DBStd	22.08	11.26	11.90	11.21	8.59	8.01	6.19	5.41	5.79	7.58	8.99	10.99	9.88		
	HDD50	3184	857	705	356	56	6	0	0	0	2	105	424	673		
	HDD65	6247	1322	1124	799	345	86	5	1	3	94	466	864	1138		
	CDD50	3761	0	0	22	171	494	767	942	799	445	110	11	0		
	CDD65	1350	0	0	0	10	109	322	478	337	88	6	0	0		
	CDH74	17782	0	0	0	171	1312	4098	6611	4299	1183	108	0	0		
	CDH80	8883	0	0	0	47	477	2012	3721	2155	453	18	0	0		
	WSAvg	3.5	2.9	3.4	4.1	5.0	4.3	3.5	3.2	3.3	3.0	2.9	2.8	3.4		
	PrecAvg	8.4	0.7	0.8	1.1	1.4	0.9	0.4	0.2	0.1	0.3	0.7	0.9	0.9		
(15) Precipitation	PrecMax	12.0	2.6	2.6	2.0	3.7	2.4	1.1	1.1	1.1	1.1	2.4	2.2	2.4		
	PrecMin	5.0	0.0	0.1	0.1	0.0	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.1		
	PrecStd	1.9	0.5	0.6	0.5	1.1	0.6	0.3	0.3	0.2	0.3	0.6	0.6	0.7		
	PrecStd	1.9	0.5	0.6	0.5	1.1	0.6	0.3	0.3	0.2	0.3	0.6	0.6	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.5	55.6	69.4	85.8	93.5	101.3	104.1	100.0	95.3	82.4	68.2	57.8		
		MCWB	42.6	48.1	53.9	61.8	63.8	68.1	68.7	64.8	64.3	58.7	51.7	46.2		
	2%	DB	43.7	47.6	64.3	78.9	89.2	97.1	100.2	97.0	90.0	76.8	61.1	50.4		
		MCWB	38.6	41.2	50.8	58.5	62.6	65.3	66.7	64.3	62.2	55.8	48.9	43.4		
	5%	DB	39.2	43.0	59.7	73.1	85.5	93.9	97.5	94.5	85.5	71.9	55.5	45.3		
		MCWB	35.8	38.1	48.6	55.7	61.0	63.6	65.8	63.6	59.9	53.3	47.1	39.7		
	10%	DB	35.6	38.6	55.2	68.3	81.5	90.6	94.8	91.0	81.1	67.1	51.0	41.1		
		MCWB	32.9	34.4	45.9	53.3	59.5	62.4	64.9	62.3	57.8	50.9	44.2	36.9		
	0.4%	WB	43.8	47.3	56.0	64.3	68.3	70.1	73.4	68.2	66.2	59.7	53.8	46.7		
		MCDB	49.1	54.5	64.6	81.8	85.9	92.9	96.0	90.3	93.0	80.1	64.5	55.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.9	41.8	52.4	60.6	65.2	67.0	69.3	65.9	63.0	56.4	50.6	44.1		
		MCDB	42.6	46.7	62.3	74.2	83.3	91.7	92.7	90.8	87.4	74.5	59.9	50.2		
	5%	WB	36.1	38.3	49.1	57.6	63.0	65.2	67.4	64.5	60.9	54.2	47.6	40.6		
		MCDB	39.2	43.0	58.9	70.7	79.9	89.4	91.2	91.2	84.3	70.5	54.7	44.7		
	10%	WB	33.3	34.8	46.4	54.8	61.1	63.8	65.8	63.2	58.5	51.4	44.6	37.6		
		MCDB	35.5	38.4	54.2	66.4	78.0	86.9	90.2	88.0	79.3	65.7	50.9	40.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.9	16.0	18.0	21.4	23.8	24.8	25.6	26.3	27.2	25.1	17.5	13.8		
		MCDBR	15.6	20.7	25.2	29.3	29.5	28.9	29.8	30.7	32.5	31.1	25.6	21.4		
		MCWBR	11.8	14.1	13.7	13.9	11.0	9.4	9.7	10.1	13.2	15.7	15.2	14.9		
	5% WB	MCDBR	14.8	18.0	23.9	25.2	24.9	26.1	27.7	28.1	31.1	29.9	23.6	18.8		
		MCWBR	11.2	13.2	13.9	13.5	11.2	9.8	10.5	9.8	13.6	15.4	14.5	13.4		
			(d) 11.2	(e) 13.2	(f) 13.9	(g) 13.5	(h) 11.2	(i) 9.8	(j) 10.5	(k) 9.8	(l) 13.6	(m) 15.4	(n) 14.5	(o) 13.4		
(40) Clear-Sky Solar Irradiance	taub		0.359	0.395	0.444	0.468	0.455	0.444	0.433	0.398	0.376	0.407	0.401	0.352		
	taud		2.038	1.991	1.974	1.965	2.056	2.125	2.168	2.249	2.267	2.084	2.045	2.072		
	Ebn at Noon		227	241	247	253	260	264	265	271	268	235	209	216		
	Edh at Noon		35	43	50	55	52	49	46	41	37	39	35	30		
(44) All-Sky Solar Radiation	RadAvg		516	814	1215	1698	2173	2395	2358	2139	1673	1047	596	422		
	RadStd		77	94	91	168	116	97	107	69	69	88	62	46		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	+1.44	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page